



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO JVS FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

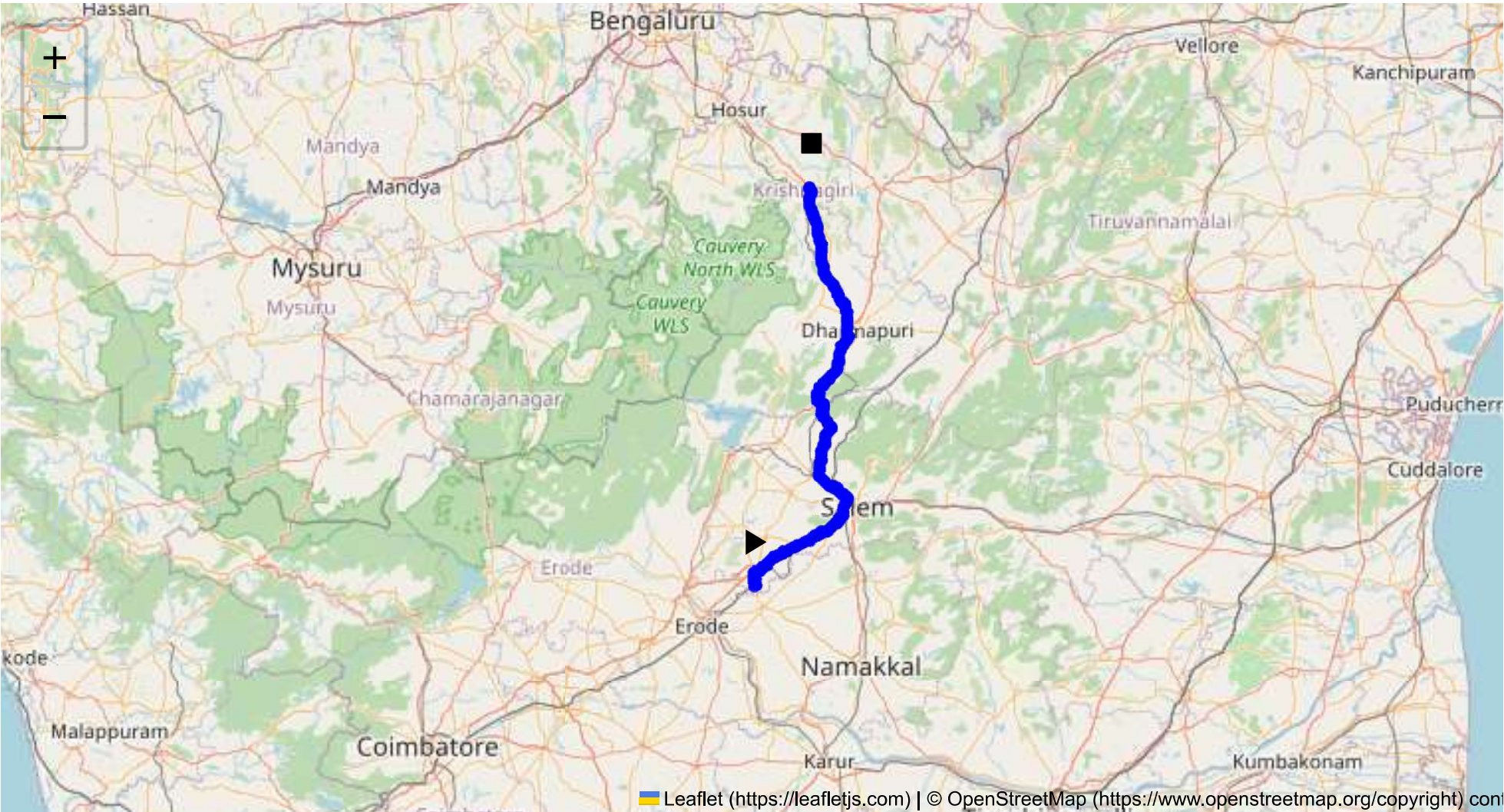
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 151.81 km
Estimated Duration: 2.8 hours
Adjusted Duration (Heavy Vehicle): 3.5 hours
Start: (11.4381, 77.8734)
End: (12.524575, 78.022536)



Welcome to the Journey Risk Management Study

Route Safety Analysis: Sangagiri to Pillari Agraharam, Tamil Nadu

- 1. **Overview of the Route Map:** The route from CVQF+23W, Sangagiri to G2FF+R3R, Pillari Agraharam spans roughly 151.81 kilometers. The primary roads involved include national highways and state highways, likely passing through urban areas, rural landscapes, and some hilly sections.

2. **Typical Weather Conditions and Potential Weather-Related Hazards:** Tamil Nadu experiences a tropical climate. From June to September, the southwest monsoon can result in heavy rains, leading to potential flooding and slippery roads. The northeast monsoon from October to December also poses similar risks. In contrast, the summer months can produce high temperatures causing road surface wear, particularly in rural areas.
3. **Traffic Patterns and Congestion:** The route likely sees increased traffic during morning (7-10 AM) and evening (5-8 PM) rush hours. Major towns along the route could experience congestion throughout the day. Factors such as local markets, festivals, or trucking hubs can also contribute to traffic buildup.
4. **Assessment of Road Quality and Infrastructure:** National highways are likely to be well-maintained, but state roads may vary in quality. Some rural roads may be narrow or have potholes, requiring caution. Bridge and overpass conditions are critical, especially during monsoon seasons.
5. **Alternative Routes for Emergencies:** In case of roadblocks or hazards, alternative state highways or local roads should be considered. Consult a real-time traffic navigation app to find viable detours. Consider routes that bypass major towns to escape congestion.
6. **Local Regulations Affecting Hazardous Material Transport:** Transporting hazardous materials might require permits, specific timing restrictions (avoiding peak hours), and adherence to maximum speed limits. Ensure compliance with Tamil Nadu's regional safety regulations and emergency protocols.
7. **Historical Incidents Involving Heavy Vehicles:** Many incidents in Tamil Nadu often involve overturned vehicles due to speeding or cornering. Incidents near industrial areas highlight the importance of speed regulation and proper vehicle maintenance. Past monsoon seasons have seen road collapses and landslides affecting routes.
8. **Environmental Considerations and Sensitive Areas:** The route could cross agricultural zones and small water bodies. Maintain caution to avoid spills or accidents which could have serious environmental impacts. Sensitive areas such as wildlife sanctuaries should be noted for reduced speed and noise regulations.
9. **Communication Coverage:** Most areas on the route will have good mobile network coverage. However, rural regions or hilly areas might experience temporary dead zones. Keep an emergency communication device or plan for alternative contact methods.
10. **Estimated Emergency Response Times:** Proximity to urban centers implies faster emergency response, typically within 30-60 minutes. Rural and remote areas might see extended response times, up to several hours depending on severity and accessibility.
11. **Overall Summary of Risk Assessment:** The route from Sangagiri to Pillari Agraharam poses moderate risk for truck drivers transporting hazardous materials. Key concerns include weather impacts, road quality, congestion, and accident risk. Proper planning, adherence to regulations, and situational awareness are vital. Alternative routes should be planned in advance, and constant communication preparedness is crucial.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.45309, 77.87325	30 KM/Hr
5	Turn	Medium	11.46400, 77.87323	30 KM/Hr
6	Turn	Medium	11.46765, 77.87037	30 KM/Hr
7	Turn	High	11.47403, 77.86841	15 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.53512, 77.95717	30 KM/Hr
11	Turn	Medium	11.53606, 77.95869	30 KM/Hr
12	Turn	Medium	11.62403, 78.10893	30 KM/Hr
13	Turn	High	11.64183, 78.11810	15 KM/Hr
14	Turn	Medium	11.64338, 78.11998	30 KM/Hr
15	Turn	Medium	11.73968, 78.05098	30 KM/Hr
16	Turn	Medium	11.86790, 78.08347	30 KM/Hr
17	Turn	Medium	11.91357, 78.05868	30 KM/Hr
18	Turn	Medium	11.92556, 78.06499	30 KM/Hr
19	Turn	Medium	11.93694, 78.05425	30 KM/Hr
20	Turn	Medium	11.95531, 78.04698	30 KM/Hr
21	Turn	Medium	11.95854, 78.04990	30 KM/Hr
22	Turn	Medium	11.96068, 78.04960	30 KM/Hr
23	Turn	Medium	11.96370, 78.05142	30 KM/Hr
24	Turn	Medium	11.96433, 78.05113	30 KM/Hr
25	Turn	High	11.96542, 78.04938	15 KM/Hr
26	Turn	High	11.96650, 78.04927	15 KM/Hr
27	Turn	Medium	12.12036, 78.13284	30 KM/Hr
28	Turn	Medium	12.50466, 78.03128	30 KM/Hr
29	Turn	High	12.51373, 78.03208	15 KM/Hr
30	Turn	Medium	12.51395, 78.03172	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
31	Turn	Medium	12.51382, 78.02925	30 KM/Hr
32	Turn	Medium	12.51628, 78.02574	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
13	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
16	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
20	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
21	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
22	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
23	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
25	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
26	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
27	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
28	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
29	hospital	Global Medical Centre Jagirammalayam	11.6894409, 78.1108847	30 km/h	Medium
30	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
33	pharmacy	MJ Medicals	11.7232728, 78.0698965	30 km/h	Medium
35	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
36	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
38	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
40	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
41	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
42	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots